TITLE: THERMO-INTENSIFIER :A novel sensor to intensify the monitoring of temperature

ABSTRACT

Background:

Heat generation and transfer to the dentine-pulp complex during various dental procedures can indeed pose risks to the health of the pulp tissue. The factors contributing to this heat transfer include friction during cavity preparations, exothermic reactions during the polymerization of restorative materials, and polishing of restorations.In vitro studies have indicated that detrimental effects may occur when the intra-pulpal temperature increases by more than 5.5°C, surpassing a threshold of 42.4°C. Excessive heat can lead to inflammation and necrosis of the pulp tissue, which can compromise the vitality of the tooth.To quantify the significance of heat transfer and control during dental procedures, experimental setups have been utilized. These setups typically involve placing a thermocouple inside the pulp chamber of an extracted human tooth and connecting it to an electronic digital thermometer. This allows researchers to monitor and measure the temperature changes occurring within the pulp tissue in response to various dental procedures.Despite the recognized importance of controlling heat transfer during dental interventions, there remains a limited number of studies that have comprehensively quantified its significance. Further research in this area is necessary to better understand the effects of heat generation on pulp health and to develop strategies for mitigating potential risks during dental treatments.

Thermo intensifier Working principle:

Our innovative and newly designed "THERMO-INTENSIFIER" appears to be a promising solution in the field of dental drilling procedures which includes tooth preparation, implant osteotomy site preparation, cavity preparation and denture fabrication. By integrating a any temperature detecting sensor (both contact and non contact) into the dental handpiece, it offers real-time temperature monitoring during drilling, which addresses a significant concern in traditional drilling methods – excessive heat generation and its associated complications.This will alter the cooling effect of handpiece (internal as well as

external coolant) based on the temperature detected. The ability to monitor temperature in real-time can provide dentists with valuable feedback, allowing them to adjust their technique to prevent thermal damage effectively. This not only enhances patient safety and comfort but also has the potential to improve treatment outcomes. Additionally, by reducing the risk of thermal damage, the device may help extend the lifespan of dental burs, resulting in cost savings for dental practices.The name "THERMO-INTENSIFIER" suggests the device's capability to intensify the monitoring of temperature using digitalised and computerised during dental procedures, which aligns well with its purpose. This innovation has the potential to redefine the safety and efficiency standards in dental procedures, making them more predictable and less prone to complications.

References:

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2. Center for Devices and Radiological Health (2017) Letter to electric dental handpiece manufacturers regarding reports of overheating and patient burns: notice of safety concern for electric dental handpieces and accessories. U.S. Food And Drug Administration. http://www.fda.gov/MedicalDevices/ResourcesforYou/ Industry/ucm226986.htm.